NEXA

TECHNOLOGY FOR A BRIGHTER TOMORROW

TECHNICAL SPECIFICATIONS

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NexaBook AI Studio 16

Technical Specifications

Built for AI Developers, Creators and Researchers



- AI-Ready Performance
- Workstation-Class Graphics
- Stunning 16-inch OLED Display
- Designed for Productivity

Push the Limits of What's Possible

NexaBook AI Studio 16 delivers uncompromising performance for local AI, machine learning, content creation and professional workloads.

AMD Ryzen™ AI 9 Processor	NVIDIA® GeForce RTX™ 5080	64 GB DDR5 Memory	16" 3.2K OLED
Up to 50 TOPS NPU	16 GB GDDR7	(up to 128 GB)	120 Hz
			100% DCI-P3

1. Product Positioning

NexaBook AI Studio 16 is Nexa's flagship mobile AI workstation for local inference, generative AI development, machine learning experimentation, GPU-accelerated creation and demanding software engineering. It prioritizes GPU memory, system memory, sustained cooling and display quality over ultra-light portability.

Core configuration

Model	NB-AIS16-2026
Processor	AMD Ryzen AI 9-class processor, 12 cores / 24 threads
NPU	Integrated AI engine, up to 50 TOPS
Graphics	NVIDIA GeForce RTX 5080 Laptop GPU, 16 GB GDDR7
Display	16" 3.2K OLED, 120 Hz, 100% DCI-P3, HDR
Memory	64 GB DDR5; expandable to 128 GB
Storage	2 TB PCIe 4.0 NVMe SSD; second M.2 slot; up to 8 TB total
Wireless	Wi-Fi 7, Bluetooth 5.4
Ports	2 x USB4, 2 x USB-A, HDMI 2.1, 2.5 GbE via adapter, SD Express, audio
Battery	99 Wh
Weight	Approx. 2.25 kg
OS	Windows 11 Pro

AI workload profile

Workload	Suitability
Cloud AI/API engineering	Excellent
RAG and embedding development	Excellent
Local 7B-14B quantized inference	Excellent
Larger local models	Good; feasibility depends on quantization, context and mem
Computer vision prototyping	Excellent
Fine-tuning / training experiments	Suitable for small-to-medium development experiments; not
Docker / local Kubernetes	Excellent

2. Thermal, Power & Expansion

The AI Studio 16 uses a dual-fan vapor-chamber cooling design intended for sustained CPU/GPU workloads. Maximum performance is available when connected to the supplied high-wattage adapter. Performance modes may trade acoustic output and battery life for sustained compute.

Memory and storage

Two serviceable memory channels support configurations up to 128 GB. Dual M.2 NVMe slots support separate OS/project drives or expanded local model storage. Exact supported capacities depend on qualified components.

3. AI Software Compatibility

Capability	Notes
CUDA-oriented development	Supported through NVIDIA GPU software stack
Local model runtimes	Suitable for common Windows-compatible local inference runtimes
Python ML environments	Supported; use isolated environments and compatible GPU drivers
.NET AI development	Suitable for Microsoft.Extensions.AI, Agent Framework, local/containerized services
Containers	Hardware acceleration depends on runtime and host configuration

Configuration disclaimer

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